

CompEng 2DX3

Final Deliverable Report

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As a future member of the engineering profession, the student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University and the Code of Conduct of the Professional Engineers of Ontario. Submitted by [Keerthna Manivasagam, 400579618]

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1. Device Overview

1A. Features

The system provides a simple two-button interface, two status LEDs, and an automated 3D scanning pipeline. The core components and their features are as follows:

Texas Instruments MSP432E401Y Microcontroller:

- ARM Cortex-M4F processor core with Floating-Point Unit (FPU)
- 33.33 MHz bus speed (assigned: 34 MHz, LSD = 8, achieved via SYSDIV2 = 5)
- I2C interface for ToF sensor communication at 100 kbps
- UART interface for PC serial communication at 115200 bps
- Onboard user LEDs: PN0, PN1, PF0, PF4
- Onboard user switches: PJ0, PJ1

VL53L1X Time-of-Flight (ToF) Sensor:

- Maximum accurate ranging distance of 4 m (long distance mode selected)
- Maximum ranging frequency of 50 Hz
- I2C interface, 2.6 V – 3.5 V operating voltage
- 940 nm infrared VCSEL emitter with SPAD detection array

28BYJ-48 Stepper Motor with ULN2003A Driver Board:

- 2048 steps per 360-degree rotation (half-step mode)
- 5 – 12 V supply voltage on driver board
- Controlled via PH0–PH3 GPIO outputs

Serial Communication:

- I2C protocol (100 kbps) for ToF sensor to microcontroller communication
- UART protocol (115200 bps) for microcontroller to PC communication

3D Visualization:

- MATLAB used for serial data reception, polar-to-Cartesian conversion, and 3D plot generation
- Scatter plot combined with ring outlines and inter-scan wireframe connections

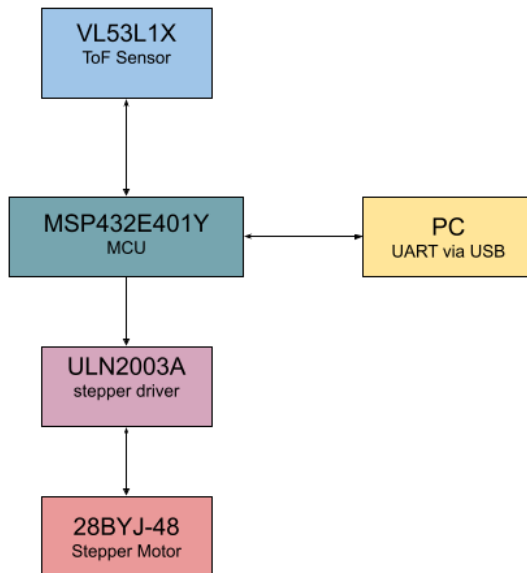
1B. General Description

The system functions as a budget-friendly embedded LiDAR room scanner which creates three-dimensional spatial maps of indoor spaces. The microcontroller MSP432E401Y serves as the central hub which manages all system operations by controlling stepper motors and gathering ToF sensor data and displaying LED status and sending UART data to a connected PC.

The system starts by initializing all connected devices and waits for the ToF sensor to finish its startup process which is shown by the activation of all four onboard LEDs. The user starts each 360-degree rotation by pressing the SW1 button which is located at PJ0. The stepper motor moves in a clockwise direction for 2048 steps while taking distance measurements from the ToF sensor every 64 steps which equals 11.25 degrees. At each measurement point the PN1 measurement status LED turns on to show that data transmission has started through UART while PF4 UART Tx status LED shows the same. The PN0 additional status LED stays illuminated during motor operation and it turns off when the motor stops.

The motor returns to its starting position at 0 degrees after it completes one full rotation which requires 32 measurements. The system indicates that homing has ended through all four LEDs which flash to show that the system has reached operational status for its next scan. The user presses SW1 again to initiate subsequent scans. The 3D plot generation executes automatically after the completion of 3 scans which result in total measurements of 96.

1C. Block Diagram



2. Device Characteristics

Feature	Detail
Microcontroller	MSP432E401Y
Bus Speed	33.33 MHz (SYSDIV2 = 5, assigned 34 MHz)
Measurement Status LED	PN1 (flashes each ToF reading)
UART Tx Status LED	PF4 (flashes at start of each transmission)
Additional Status LED	PN0 (on while motor running)
Start/Stop Button	PJ0 (SW1, onboard, active-low)
Baud Rate	115200 bps
Stepper Motor	28BYJ-48 + ULN2003A Driver
IN1	PH0
IN2	PH1
IN3	PH2
IN4	PH3
VCC	5V
GND	GND
ToF Sensor	VL53L1X
VIN	3.3V
GND	GND
SDA	PB3
SCL	PB2
I2C Speed	100 kbps
Measurements per scan	32 (every 11.25 degrees)
Steps per rotation	2048 (half-step mode)
ToF Distance Mode	Long distance (up to 4 m)
ToF Timing Budget	200 ms

3. Detailed Description

3A. Distance Measurement

The VL53L1X Time-of-Flight sensor measures distance by emitting an infrared light pulse and detecting the reflected signal. The distance calculation uses the time delay between emission and reception with the following formula:

$$D = \frac{c \cdot t}{2}$$

In this case, c is the speed of light (299 792 458 m/s) and t is the travel time. The sensor uses its internal system to conduct optical emission detection while it processes signals through its signal conditioning system and its analog-to-digital converter. The system produces a digital output that measures distance in millimeters, which it sends to the microcontroller through I2C. The system conducts a total of 32 measurements during each complete rotation, which results in angular measurements that divide into the following intervals:

$$\frac{360^\circ}{32} = 11.25^\circ$$

The stepper motor operates with 2048 steps for each complete revolution which means,
 $\frac{2048}{32} = 64 \text{ steps per measurement}$

This info is integrated into the Keil software to determine the frequency of distance measurements. The system acquires distance measurements in polar form which consists of distance and angle measurements. The system uses MATLAB to convert these polar measurements into Cartesian coordinates (see below).

$$Z = d \sin\theta, Y = d \cos\theta, X = 100\text{mm} \cdot \text{scan index}$$

It takes place after the UART transmission, allowing the PC to perform highly accurate calculations instead of the microcontroller. The process of transforming 2D scans into 3D models needs displacement as a fundamental requirement. The system operates without physical movement because it uses simulated displacement instead. The system assigns new x-values to each complete scan which creates a new layer while maintaining 100 mm distance between layers. The method creates a 3D model without needing advanced hardware because it operates with fundamental requirements.

3B. Visualization

The data analysis and visualization process uses MATLAB as its primary tool. The workflow requires serialport() function to establish serial communication, readline() function to read incoming data, sscanf() function to parse values. The system uses arrays to store data and cosd() and sind() functions to convert coordinates. The system uses scatter3() and plot3() functions to create plots. The final visualization displays a 3D wireframe model which contains circular scan rings and vertical connections that link to different scans. The representation establishes an unambiguous spatial distribution of the surroundings.

4. Application Note, Instructions, and Expected Output

4A. Real-World Applications

Robotic navigation and autonomous drone path planning and indoor mapping for facility management and archaeological surveying all benefit from the widespread use of this embedded LiDAR system. The system's low cost and compact design make it ideal for situations where commercial LiDAR equipment would be too expensive or too large to use.

4B. Setup and Usage Instructions

This setup assumes Keil uVision and MATLAB are installed and that the TM4C1294 LaunchPad is connected via USB.

1. Determine the COM port: open Device Manager, expand Ports (COM & LPT), and note the port listed as XDS110 Class Application/User UART.
2. Open the MATLAB script Deliverable_2_400579618.m and update the port variable on line 25 to match the COM port identified in step 1.
3. Wire the circuit according to the schematic in Section 6: ToF VIN to 3.3V, GND to GND, SDA to PB3, SCL to PB2; driver board IN1-IN4 to PH0-PH3, VCC to 5V, GND to GND.
4. In Keil, open the project, click Translate , Build, and Load to flash the firmware.
5. Press RESET on the board. All four onboard LEDs will flash briefly, confirming the ToF sensor has booted successfully.
6. Run the MATLAB script. When prompted, press Enter in the MATLAB command window.
7. Press SW1 (PJ0) on the board to start the first scan. PN0 turns on (motor running), PN1 flashes at each of the 32 measurement points, and PF4 flashes at each UART transmission.
8. After one full rotation, the motor automatically homes CCW back to 0 degrees. All LEDs flash when homing is complete.
9. For Deliverable 2 demonstration: keep the sensor at the same position and press SW1 two more times to complete scans 2 and 3. The code treats the three scans as 10 cm apart.
10. After all 96 measurements are received, MATLAB automatically generates and displays the 3D plot. The raw data is also saved to a timestamped CSV file.

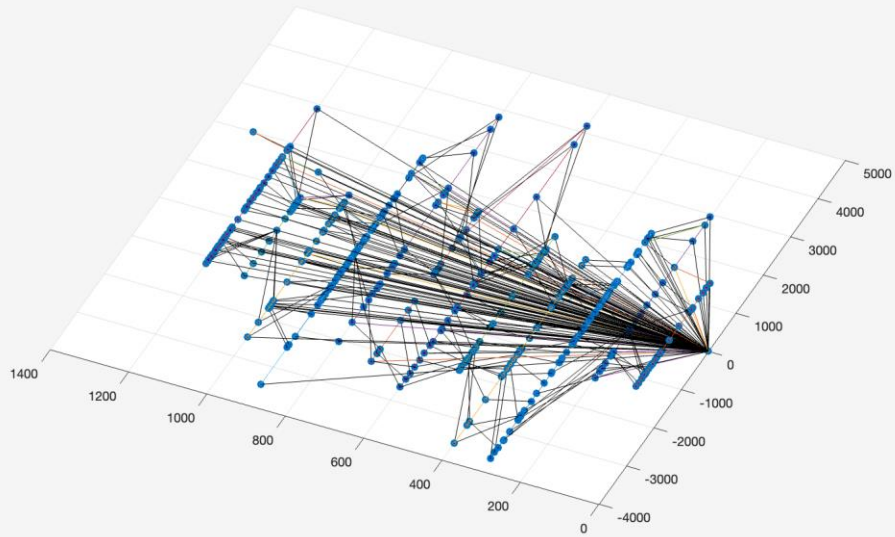
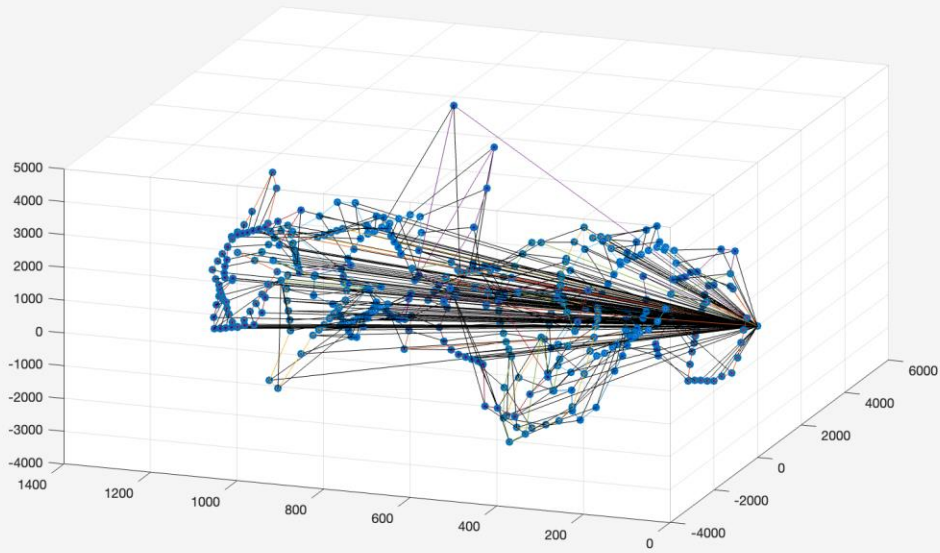
4C. Expected Output

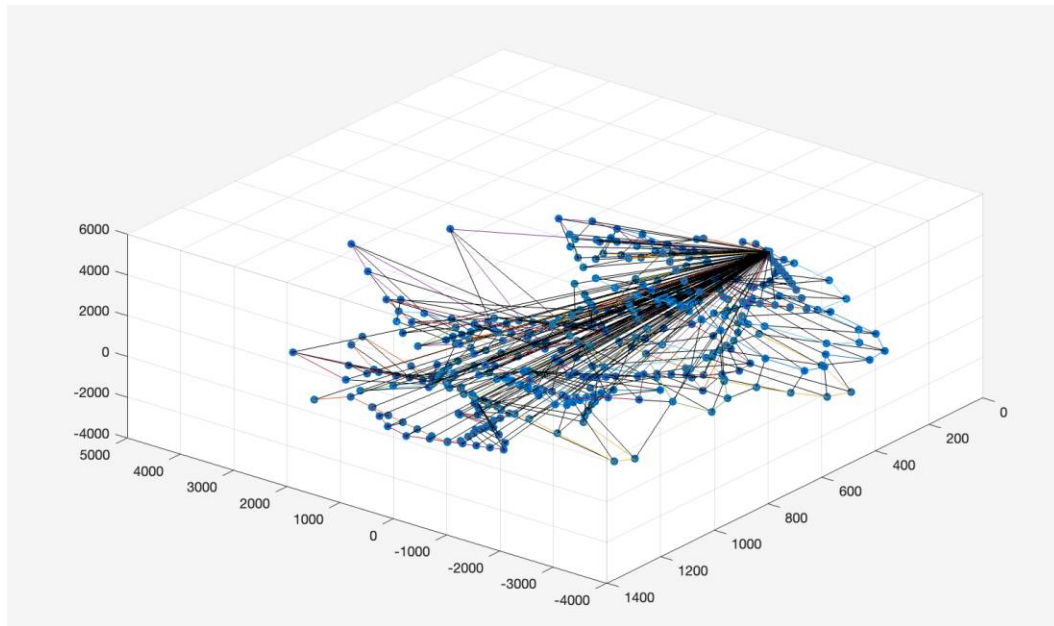
Assigned Campus Location: J - JHE Second Floor (see below)



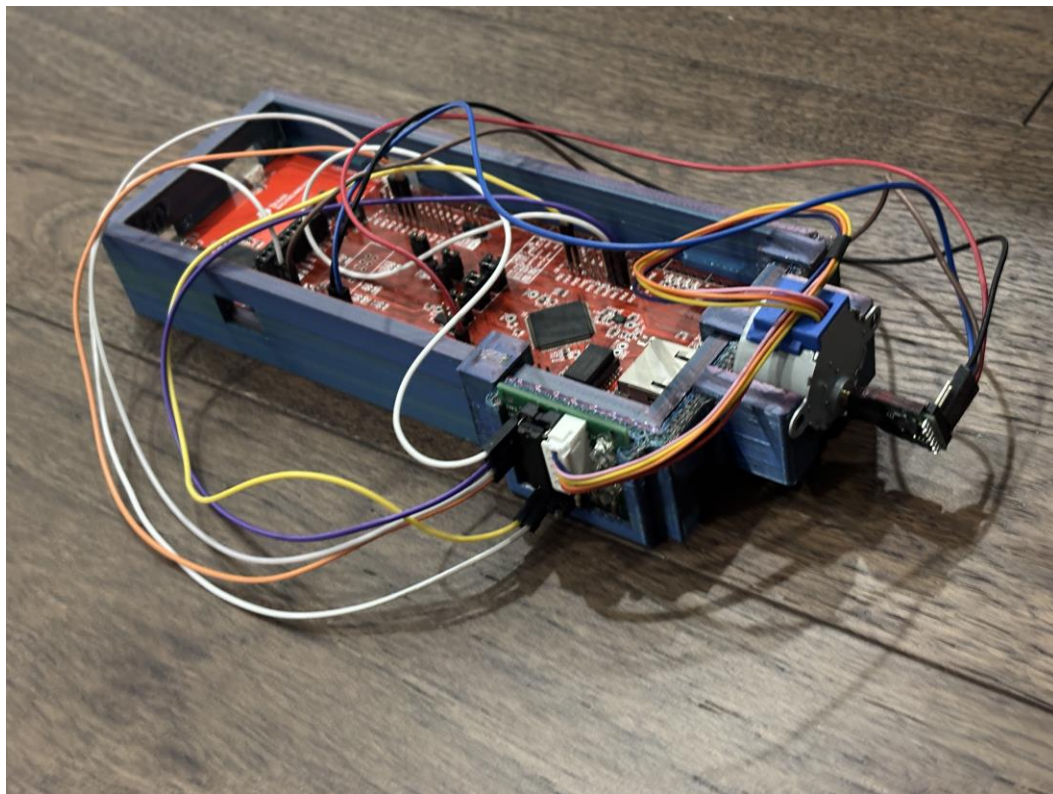
This location has multi-level ceilings, windows, and open space on the left.

Obtained Scan (Multiple Views):





Physical Device Image:



5. Limitations

5.1 Floating-Point Capability and Trigonometric Functions

The MSP432E401Y microcontroller has a Floating-Point Unit which operates at single-precision with 32-bit processing capabilities. The system handles basic arithmetic operations correctly but fails to deliver accurate results during trigonometric calculations which include sine and cosine functions.

Trigonometric functions produce irrational values that cannot be exactly represented in binary floating-point formats. This leads to the introduction of rounding errors which occur during the calculation process. The errors start from the process of converting polar coordinates which include distance and angle into Cartesian coordinates, and these errors continue to spread through all subsequent calculations.

The reconstructed 3D model contains small inaccuracies because three factors interact together. The three factors include restricted floating-point precision, the continuous execution of trigonometric calculations, and the aggregation of rounding mistakes. The resulting output shows three types of errors which include small undulations, improper point distribution and shifted scan layer alignment. The system achieves better accuracy through double-precision coordinate transformations which MATLAB executes instead of using embedded computation.

5.2 Maximum Serial Communication Rate

The PC supports a maximum standard serial communication rate which reaches 128000 bps. The system selected a baud rate of 115200 bps because it provided reliable communication. The value was experimentally validated through three methods which included checking COM port settings in Windows Device Manager, testing all supported baud rate options of the USB-to-UART interface, and testing real-time data transmission in MATLAB. The system maintained stable operation at 115200 bps while transmitting data without any loss or corruption, which proved that this rate operated within the system's dependable performance range.

5.3 MCU and ToF Sensor Communication

The microcontroller communicates with the VL53L1X Time-of-Flight sensor through the I2C protocol which operates as a synchronous two-wire interface. The I2C bus operates at approximately 100 kHz, as configured through the I2C timing register. The ToF sensor itself supports measurement rates up to 50 Hz, but in this system, the effective sampling rate is reduced due to configuration choices such as a 200 ms timing budget per measurement. The system experiences its main performance limitations through measurement timing, which establishes itself as the dominant factor, despite I2C offering dependable communication between devices.

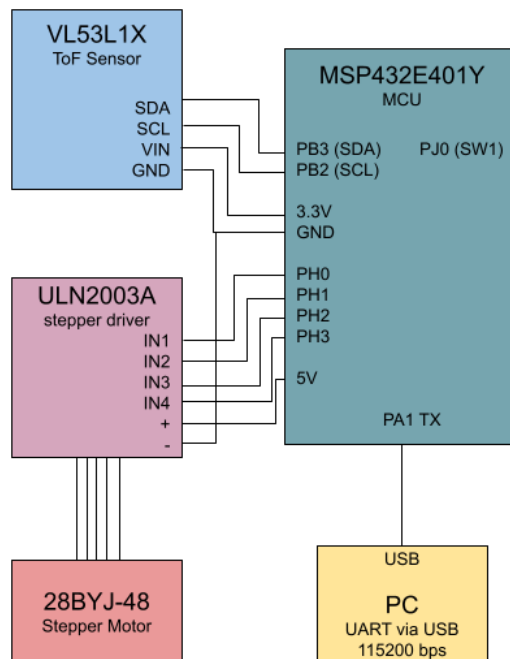
5.4 System Speed Verification and Limitations

The most important restriction exists because the sensor requires a 200-millisecond time budget that permits only five measurements per second. The system gains better measurement stability from this trade-off while losing its ability to respond quickly to changes. The stepper

motor needs specific time intervals between its different operational states to avoid stalling. The system implements clockwise rotation through a step delay of approximately 2 milliseconds while it uses a 4-millisecond delay for counterclockwise movement during homing. The system needs 2048 steps to complete one rotation which results in significant time delays. The system implements a polling loop to determine data readiness which creates additional waiting periods because the microcontroller continuously checks the sensor instead of using interrupts.

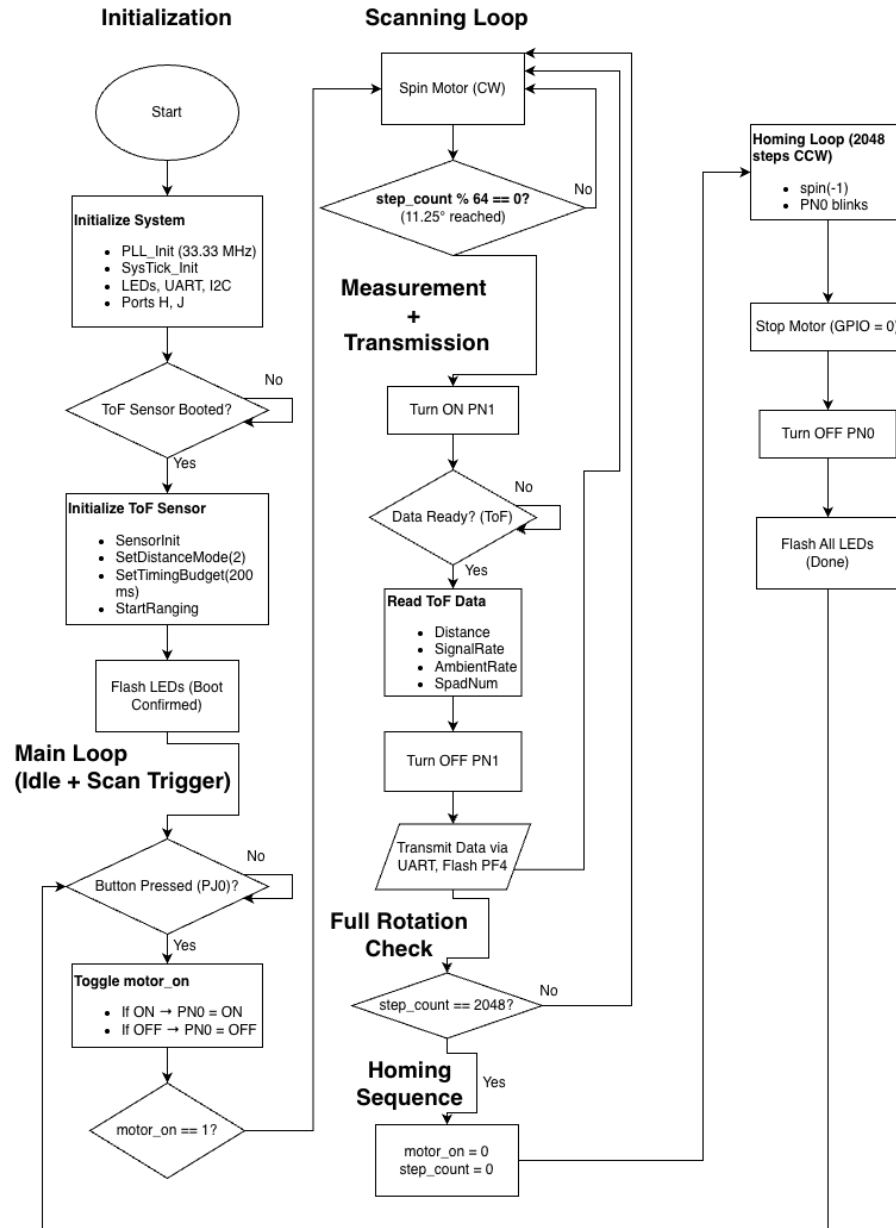
The experimental testing confirmed the verification of these limitations. The experimenters tested step delays at various increments until they reached the point of instability, which established the minimum reliable delay. The research tested various timing budgets to study how they affected the balance between speed and accuracy. The total scan duration showed a comparison with expected theoretical timing, which established that sensor timing and motor delays were the primary factors affecting performance.

6. Circuit Schematic



7. Programming Logic Flowchart

Programming Logic Flowchart for 2DX3_Project.c



Programming Logic Flowchart for 2DX3_Project.m

